1.) Suppose air is pumped into a spherical balloon.

(a) Name any physical quantities associated to the balloon which are changing.

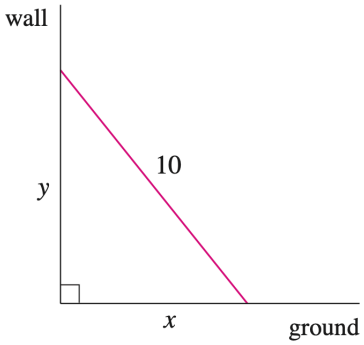
(b) Are the rates of change of these quantities independent of each other? Why or why not?

(c) Write an equation for the volume of the balloon.

(d) The volume function depends on the radius r, which itself depends on time t. Differentiate both sides of your equation with respect to t using the Chain Rule to relate $\frac{dV}{dt}$ to $\frac{dr}{dt}$.

(e) Suppose air is being pumped into the balloon so that its volume increases at a rate of 100 cm^3/s. How fast is the radius of the balloon increasing when the diameter is 50 cm?

2.) A 10 ft latter rests against a vertical wall. Let x be the distance from the bottom of the ladder to the wall and y be the distance from the top of the ladder to the ground.



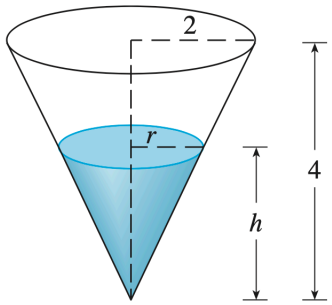
(a) Write an equation relating x and y.

(b) Differentiate each side with respect to t using the Chain Rule to relate $\frac{dx}{dt}$ to $\frac{dy}{dt}$.

(c) If the bottom of the ladder slides away from the wall at a rate of 1 ft/s, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 ft from the wall?

Name:_____

3.) A water tank has the shape of an inverted circular cone with base radius 2 m and height 4 m. If water is being pumped into the tank at a rate of 2 m^3/min, find the rate at which the water level is rising when the water is 3 m deep. (*Hint*: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$. Use the diagram below to eliminate r.)



4.) Car A is traveling west at 50 mph and car B is traveling north at 60 mph. Both are headed for the intersection of the two roads. At what rate are the cars approaching each other when car A is 0.3 mi and car B is 0.4 mi from the intersection. (*Hint:* Draw a diagram.)

Name:_____

5.) A man walks along a straight path at a speed of 4 ft/s. A searchlight is located on the ground 20 ft from the path and is kept focused on the map. At what rate is the searchlight rotating when the man is 15 ft from the point on the path closest to the searchlight?

